UT Austin EDS — Developer Guide

Date: April 2026

Adobe

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UT Austin EDS — Developer Guide

Project: University of Texas at Austin
Platform: AEM Edge Delivery Services (xwalk)
Repository: `github.com/prajwal85/eds-agent-demo`
Date: April 2026

Architecture Overview

Component	Technology
Project Type	xwalk (Universal Editor authoring)
Source CMS	Drupal 11 (migrated from utexas.edu)
Delivery	AEM Edge Delivery CDN
Author	`author-p11300-e47725.adobeaemcloud.com`
Preview	`main--eds-agent-demo--prajwal85.aem.page`
Live	`main--eds-agent-demo--prajwal85.aem.live`

Repository Structure

```
/
├── blocks/                    # Custom block implementations
│   ├── hero-video/           # Homepage video hero
│   ├── cards-article/        # News story cards grid
│   ├── columns/              # Vanilla columns (with sticky-panels + video detection)
│   ├── columns-promo/        # Promotional callout
│   ├── columns-impact/       # Impact stats sections
│   ├── columns-outro/        # Outro with video
│   ├── columns-resource/     # Resource link lists
│   ├── sticky-panels/        # Sticky scroll panels
│   ├── hero-banner/          # Interior page hero
│   ├── header/               # Site header
│   └── footer/               # Site footer
├── styles/
│   └── styles.css            # Global styles + brand tokens
├── scripts/
│   ├── scripts.js            # Core EDS scripts
│   └── delayed.js            # Deferred scripts
├── tools/importer/           # Import infrastructure
│   ├── import-homepage.js    # Homepage import script
│   ├── import-interior.js    # Interior pages import script
│   ├── import-resource-hub.js # Resource hub import script
│   ├── parsers/              # Block parsers (per-block content extraction)
│   └── transformers/         # Page transformers (cleanup, sections)
├── content/                  # Migrated content + media
│   ├── index.plain.html     # Homepage
│   ├── images/homepage/      # Homepage images
│   └── media/               # Video files
├── component-definition.json # Universal Editor block definitions
```

```
├── component-models.json     # Block field models
├── component-filters.json    # Block container filters
└── Instructions.md           # Migration reference doc
```

Custom Blocks

hero-video

Files: `blocks/hero-video/hero-video.js`, `.css`, `_hero-video.json`

Behavior: - Detects `.mp4` link in Row 2 (video field) - Creates `<video autoplay muted loop playsinline>` element - Uses Row 1 image as poster fallback - Removes video row after creating element, leaving 2 rows (background + text)

CSS Pattern: - `div:first-child` — absolute positioned background (video/image) - `div:last-child` — relative positioned text overlay with gradient

Model (3 fields = 2 cells):

```
image (reference) + imageAlt (collapsed)
video (reference)
text (richtext)
```

columns (enhanced)

File: `blocks/columns/columns.js`

The vanilla columns block has been enhanced with auto-detection:

1. **Sticky Panels Detection:** If column 1 has image AND column 2 has `<strong>` stats + `<a>` CTA → applies sticky-panels CSS classes + IntersectionObserver for fade-in
2. **Video Detection:** If any column has a link containing `.mp4` or `/media_` → creates autoplay video element
3. **Fallback:** Standard columns behavior (image-col detection)

Why: AEM Author delivers impact sections as `columns` blocks. The JS auto-detects content patterns and applies appropriate decoration.

cards-article

File: `blocks/cards-article/cards-article.css`

CSS Grid Layout (desktop):

```
┌──────────────────────┬──────────────────────┐
│ Card 2 (secondary)   │                      │
│ [img] [headline]     │  Card 1 (prominent)  │
├──────────────────────┤  [full image]        │
│ Card 3 (secondary)   │  [headline overlay]  │
│ [img] [headline]     │                      │
└──────────────────────┴──────────────────────┘
```

- `li:first-child` — right column, spans both rows, full-height image with gradient headline overlay
- `li:nth-child(2/3)` — left column, horizontal layout (image left, headline right)

sticky-panels

Files: `blocks/sticky-panels/sticky-panels.js`, `.css`

Desktop Behavior: - Image column: `position: sticky; top: 64px; height: calc(100vh - 64px)` - Content column: normal flow, scrolls past sticky image - Stats: `opacity: 0 → 1` transition via IntersectionObserver

Mobile: Stacked layout (image on top, content below)

Model: 2 cells per block (image + content). Use 3 consecutive blocks for 3 panels.

columns-outro

File: `blocks/columns-outro/columns-outro.js`

Detects `.mp4` links and creates autoplay video. Adds `.columns-outro-video-col` class for styling.

Design Tokens (styles/styles.css)

```
:root {
  /* UT Austin brand colors */
  --link-color: #bf5700;          /* Burnt orange */
  --link-hover-color: #9d4700;    /* Darker orange */
  --dark-color: #333f48;          /* Charcoal */
  --text-color: #212529;          /* Body text */
  --heading-color: #333f48;       /* Headings */
  --burnt-orange: #bf5700;        /* Brand alias */
  --charcoal: #333f48;            /* Brand alias */
  --background-color: white;
  --light-color: #f8f8f8;
  --overlay-background-color: #ebe7e1;

  /* Fonts */
  --body-font-family: roboto, roboto-fallback, sans-serif;
  --heading-font-family: roboto-condensed, roboto-condensed-fallback, sans-serif;

  /* Button style: 4px radius, orange border */
}
```

Import Infrastructure

Overview

Content was migrated from Drupal 11 (utexas.edu) using 3 import scripts:

Script	Template	Pages
`import-homepage.js`	Homepage	1
`import-resource-hub.js`	Resource hub	3
`import-interior.js`	All other pages	45

Parsers (`tools/importer/parsers/`)

Parser	Block	Source Selector

hero-video.js	hero-video	.block-coresite-homepagehero
hero-banner.js	hero-banner	.block-bundle-utexas-hero
cards-article.js	cards-article	.block-coresite-stories
columns-impact.js	columns-impact	.block-coresite-impact-1/2/3
columns-promo.js	columns-promo	#block-coretheme-homepagemidpagepromo
columns-outro.js	columns-outro	.block-homepage-outro
columns-resource.js	columns-resource	.utexas-layout--twocol-wrapper

Transformers (`tools/importer/transformers/`)

Transformer	Purpose
utexas-cleanup.js	Remove header/footer/nav, convert blockquotes to quote blocks, insert section-metadata
utexas-sections.js	Insert section breaks before parsing (beforeTransform)

Re-importing a Page

```
# Re-bundle after parser/transformer changes
SCRIPTS="/home/node/.excat-marketplace/excat/skills/excat-content-import/scripts"
"$SCRIPTS/aem-import-bundle.sh" --importjs tools/importer/import-interior.js

# Re-import single page
echo "https://www.utexas.edu/about-texas" > /tmp/single-url.txt
node "$SCRIPTS/run-bulk-import.js" \
  --import-script tools/importer/import-interior.bundle.js \
  --urls /tmp/single-url.txt
```

Known Issues & Workarounds

1. Stats Not Extracted from Impact Sections

Cause: JSDOM displaces `.stats` elements outside parent due to `<picture>` with empty `<source>` tags breaking DOM nesting. **Fix:** Parser uses `KNOWN_STATS` fallback keyed by heading text.

2. Video on AEM Cloud

Cause: Relative `./media/` paths don't resolve on AEM delivery. **Fix:** Videos use `/content/media/` absolute paths. Both `hero-video.js` and `columns.js` detect `.mp4` AND `/media_` hash URLs.

3. Sticky Panels on AEM Author

Cause: AEM Author delivers as `columns` blocks, not `sticky-panels`. **Fix:** `columns.js` auto-detects the pattern (image + stats + CTA) and applies sticky-panels decoration.

4. xwalk/max-cells Lint Rule

Cause: Maximum 4 cells per block model. **Fix:** Sticky-panels uses 2 cells (image + content) per instance. 3 consecutive blocks for 3 panels.

Development Commands

```
# Start local dev server
aem up

# Run linting
npm run lint

# Preview at
# http://localhost:3000/content/index

# Run only CSS lint
npm run lint:css

# Run only JS lint
npm run lint:js
```

Deployment

Code deploys automatically via GitHub push → AEM Code Sync:

1. Push to `main` branch
2. GitHub Actions runs lint check
3. AEM Code Sync picks up changes
4. Preview updates at `.aem.page`
5. Publish via Sidekick to go live at `.aem.live`

Key Files Reference

File	Purpose
`fstab.yaml`	AEM Author mountpoint configuration
`.migration/project.json`	Migration project settings (type: xwalk)
`component-definition.json`	Block definitions for Universal Editor
`component-models.json`	Block field models
`component-filters.json`	Block container component filters
`Instructions.md`	Complete migration reference (DOM patterns, debugging)